Nagar Yuwak Shikshan Sanstha's
Yeshwantrao Chavan College of Engineering
(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)
Hingna Road, Wanadongri, Nagpur - 441 110
NAAC A++
Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu
Department of Computer Technology

Vision of the Department

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission of the Department

To nurture computer engineering professionals by offering: Dynamic learning environment that encourages problem-solving using emerging technology, · National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry–institute collaboration, Holistic, ethical and sustainable development for lifelong success.

Session 2026-2027(ODD)

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Srushti Chaudhari



17-08-2026
Name and Signature of Student and Date
(Signature and Date in Handwritten)

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Session	**2026-27 (ODD)**	**Course Name**	**LAB: Software Engineering**
Semester	**7th**	**Course Code**	**23CT1704**
Roll No	**24**	**Name of Student**	**Srushti Chaudhari**

Practical Number	**5**
Course Outcome	Upon successful completion of the course the students will be able to; 1. Choose appropriate software engineering process model, requirement engineering principles and software designing fundamentals for a given project. 2. Select appropriate testing strategy and apply testing principles for testing a given application. 3. Apply basics of software configuration management, version control and change control in software development. 4. Evaluate cost estimation, effort and severity of software risk for given application. 5. Perform basic operations on Sub-version for software version control.
Aim	To study and draw UML Class Diagram for the given Case Study
Problem Definition	To study, analyze, and draw a **UML Class Diagram** for the Women Safety Alert System. The diagram represents the static structure of the system by identifying the major classes, their attributes, operations, and relationships. It provides a structural view of how users, emergency contacts, alerts, locations, notifications, and authorities interact within the system.
Theory (100 words)	**A UML Class Diagram is a structural diagram used to represent the static structure of a software system. It describes classes, their attributes, operations, and relationships with other classes. A class is represented using a rectangle divided into three sections: class name, attributes, and operations. UML class diagrams can represent relationships such**

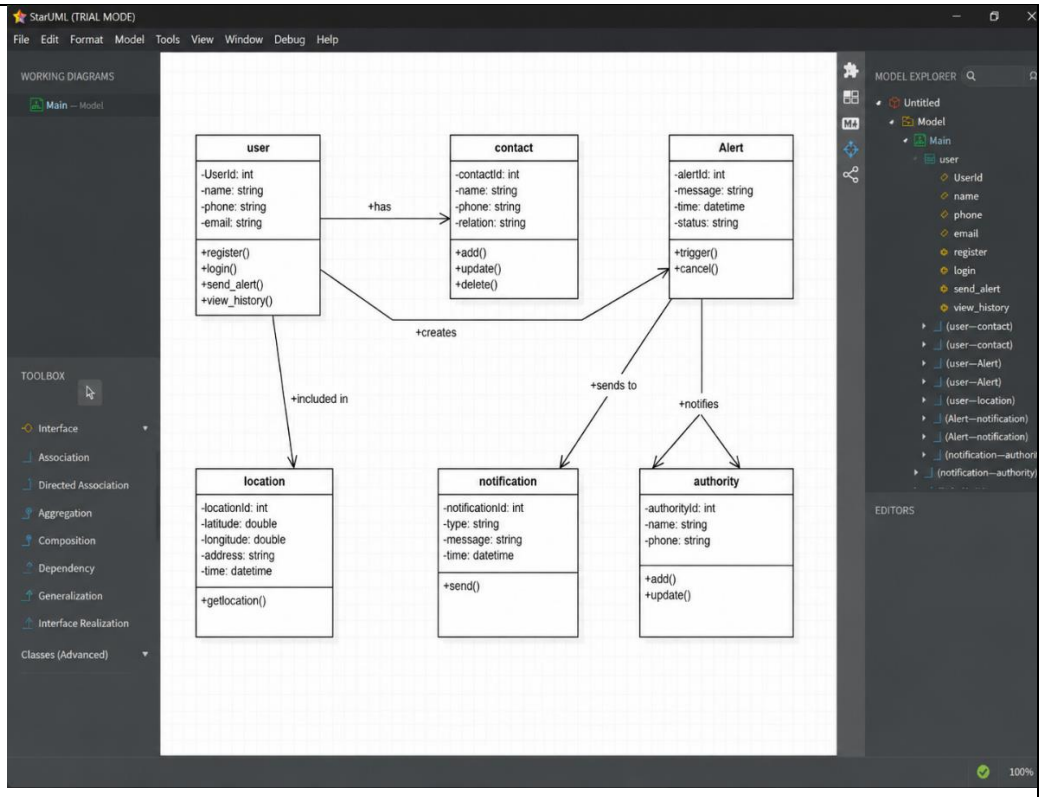
	as association, aggregation, composition, dependency, and generalization. Visibility symbols such as +, -, and # represent public, private, and protected members respectively. In the Women Safety Alert System, classes such as User, Contact, Alert, Location, Notification, and Authority are used to model the important entities and their interactions.
Procedure and Execution (100 Words)	Algorithm(Steps): **1. Start** **2. Study the case study.** **3. Identify important entities.** **4. Create classes for the identified entities.** **5. Add attributes to each class.** **6. Add operations to each class.** **7. Identify relationships between classes.** **8. Draw associations/dependencies between classes.** **9. Add relationship labels.** **10. Arrange the classes properly.** **11. Verify the diagram.** **12. Save/export the UML Class Diagram.** **13. Stop**
	Code: **Not Applicable.** **This practical involves UML modeling and graphical representation of the system structure rather than executable programming code.**
	Output:

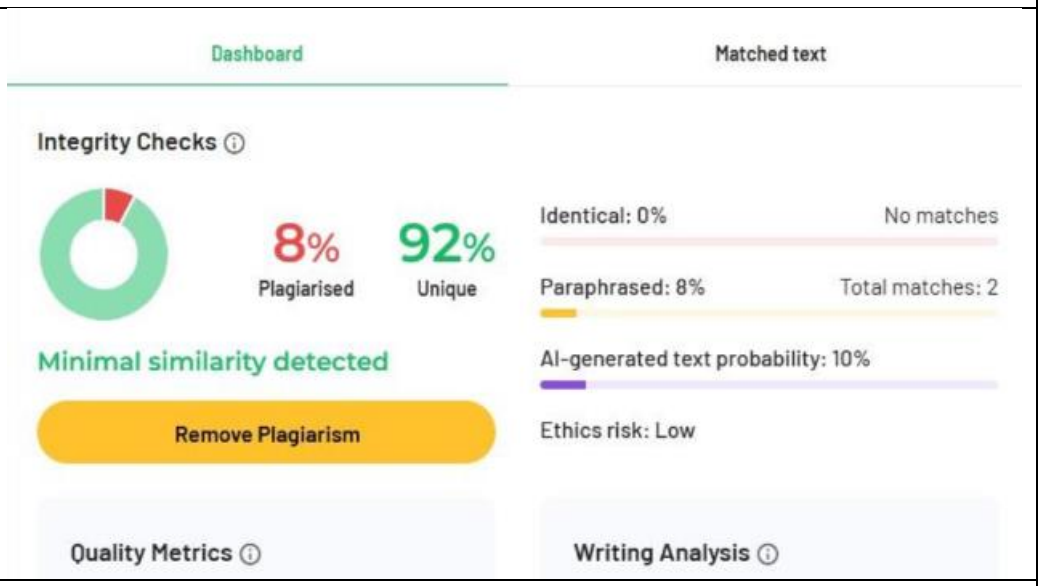
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<table>
<tr><td></td><td> 
</td></tr>
<tr><td>Output Analysis</td><td>The Class Diagram provides a structural representation of the Women Safety Alert System. The user class represents the system user, while contact stores emergency-contact information. The Alert class represents emergency alerts generated by the user. The location class stores location-related information, while notification represents messages generated from alerts. The authority class represents the emergency authority receiving notifications. The relationships between these classes show how users create alerts, maintain contacts, associate location information, and send notifications to authorities.</td></tr>
<tr><td>Link of student Github profile where lab assignment has been</td><td>https://github.com/SRchaudhari15/Software-Engineering-Practicals-24.git</td></tr>
</table>

uploaded	
Conclusion	Thus, the UML Class Diagram for the Women Safety Alert System was successfully studied and drawn using StarUML. The major classes, attributes, operations, and relationships of the system were identified and represented graphically. The Class Diagram provides a clear understanding of the static structure of the system and can be used as a foundation for system design and implementation.
Plag Report (Similarity index < 12%)	
Date	17-08-2026